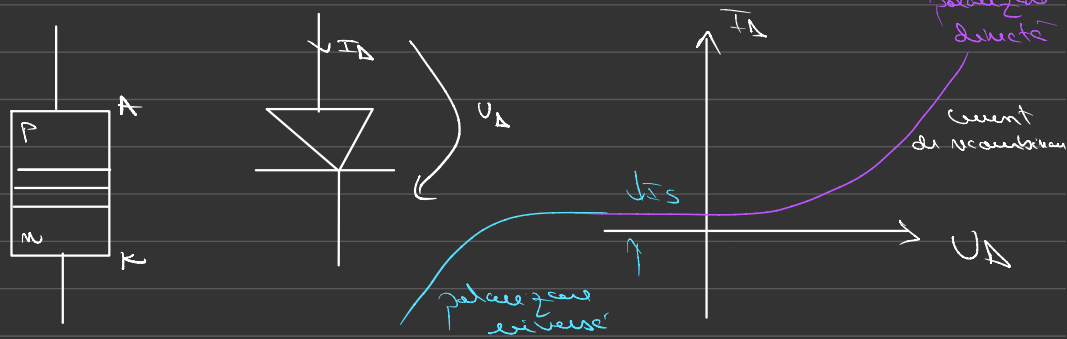
Curs 3

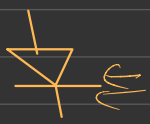


$$I_A = \begin{cases} I_S\left(e^{\frac{U_A}{U_T}} - 1\right), & U_A \geq 0 \\ -I_S, & U_A < 0 \end{cases}$$

$I_S \approx 2 \cdot 10^{-14}\ A$

TIPURI CONSTITUTIVE

1)  $\Delta \cdot O.$

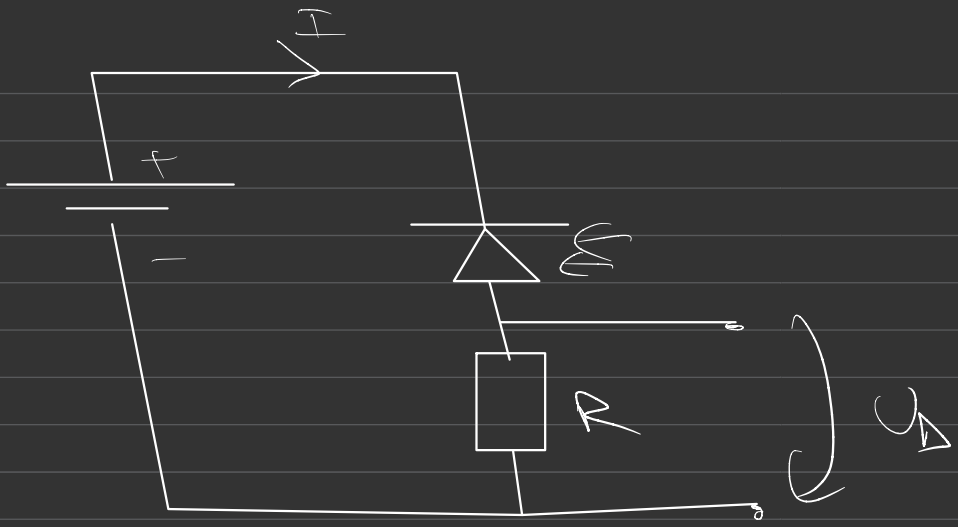
2)  fotodiodă

Pol. inversă

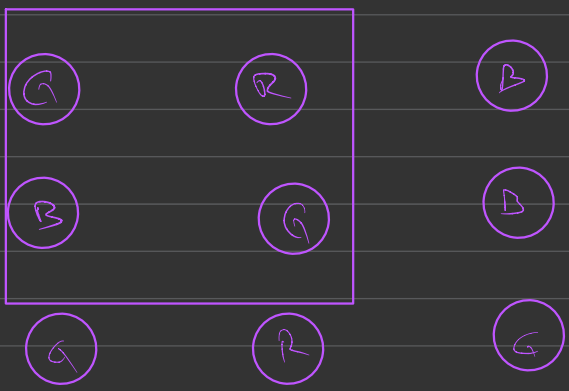
Lumină → generare

$\left(\frac{W}{m^2}\right)$

$I_A \approx E_v \cdot A$



$$U_D = I_D \cdot R = I_S \cdot R \approx E_V \cdot A/R$$

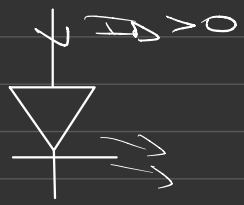


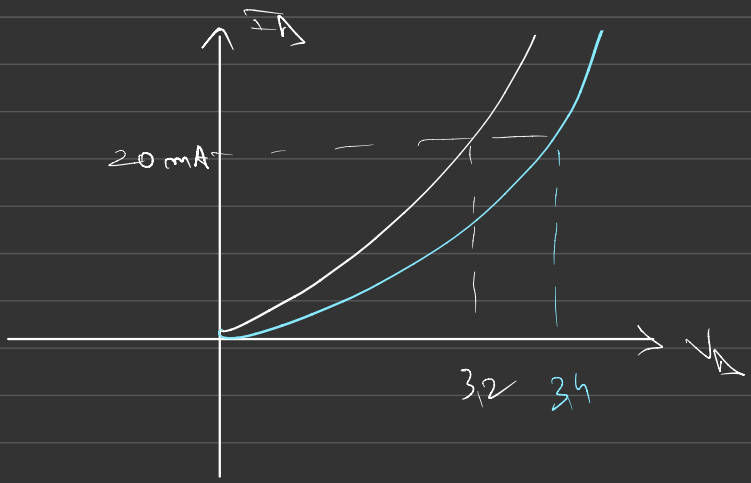
3)  L.E.D (emite lumină)

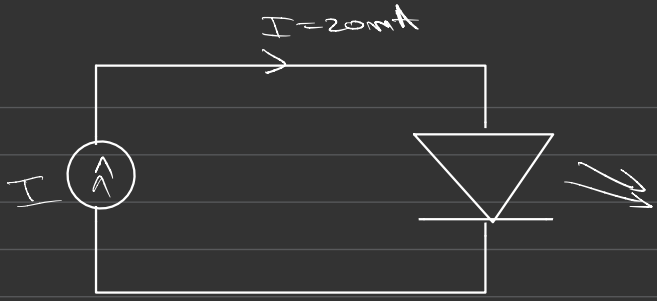
Recunoaștere

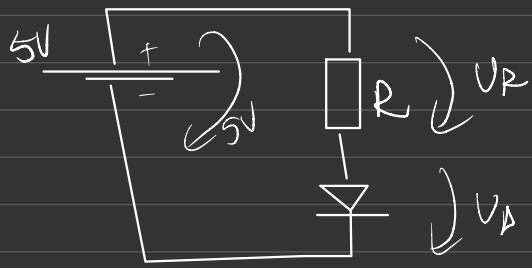
E - G → Lumină

Polarizare directă









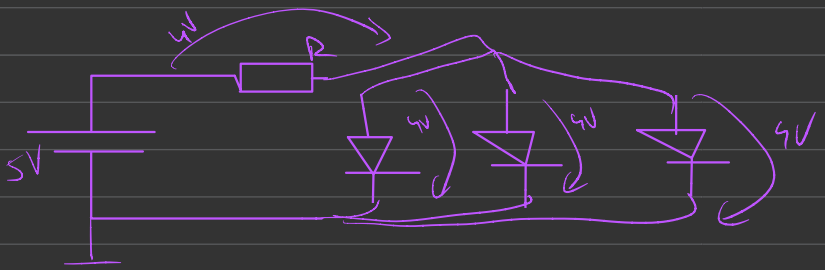
$\Rightarrow E = U_D + I_D R$

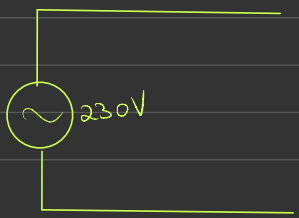
5V 3,2V 20mV

Calculăm pentru $V_{f\,max}$

$\Rightarrow R = \frac{5V - 3{,}2\,V}{20\,mA} = 0{,}09\,K\Omega$

$I_{D_{f=3,4}} = \frac{5V - 3{,}4V}{0{,}09\,K\Omega} = 17{,}7\,mA$





Modele Matematice pentru Diode

(Aproximații)



$E = U_R + U_D$

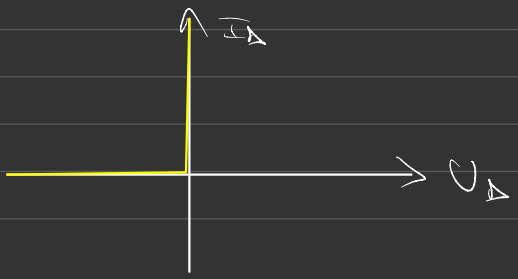
$E = IR + U_D$

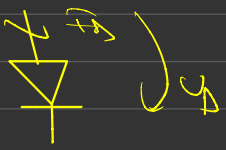
$$E = RI_S\left(e^{\frac{U_D}{U_T}} - 1\right) + U_D$$

(TRANSCENDENTALĂ)

Aprox 1

Dioda ideală

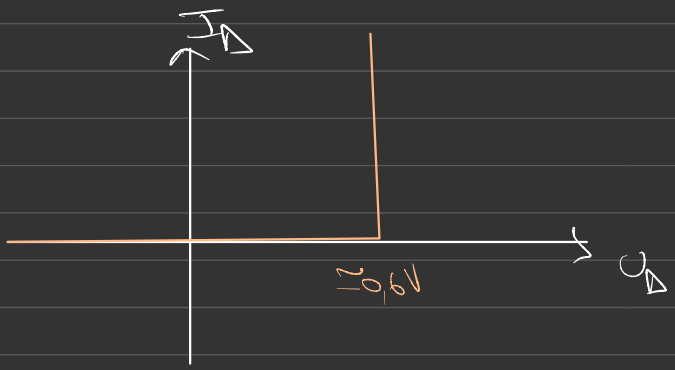




$$\begin{cases} I_D = 0 \;, & V_D < 0 \\ V_D = 0 \;, & I_D > 0 \end{cases}$$

Apax2

Tensione Constante



$$\begin{cases} I_D = 0 \;, & V_D < 0,6 \\ V_D = 0,6 \;, & I_D > 0 \end{cases}$$